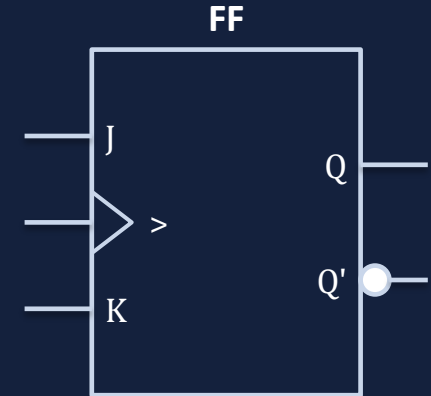


## Lecture 23

# Review for Exam 3

Latches and flip-flops · timing · sequential analysis · state reduction and design · registers and counters · practice set

Reading: Mano & Ciletti, *Digital Design with an Introduction to the Verilog HDL*, 6th ed. — Chapters 5–6 (review)



# Today's roadmap

1

§5.4

## Storage elements

Lectures 18–19

2

§5.5

## Analysis

Lecture 19

3

§5.8

## Reduction, design

Lecture 20

4

§6.4

## Counters

Registers, counters: 21–22

## By the end of class you can...

- Draw latch and flip-flop outputs from a timing diagram
- Compute maximum clock frequency and check hold
- Analyze a clocked circuit to a state table and diagram
- Reduce states and design with D, JK or T flip-flops
- Trace shift registers and design counters

# Agenda for today's session

|       |         |          |       |        |          |          |      |
|-------|---------|----------|-------|--------|----------|----------|------|
| Intro | Storage | Analysis | Break | Design | Counters | Practice |      |
| 0:00  | 0:05    | 0:25     | 0:45  | 0:55   | 1:20     | 1:35     | 1:50 |

| Elapsed time | Topic                            | In-class work |
|--------------|----------------------------------|---------------|
| 0:00 – 0:05  | Welcome; what the exam covers    | —             |
| 0:05 – 0:25  | Latches, flip-flops, timing      | Reviews 1–3   |
| 0:25 – 0:45  | Sequential analysis              | Review 4      |
| 0:45 – 0:55  | Break                            |               |
| 0:55 – 1:20  | State reduction and design       | Reviews 5, 6  |
| 1:20 – 1:35  | Registers and counters           | Reviews 7, 8  |
| 1:35 – 1:50  | Timed practice set and solutions | Practice set  |

# What Exam 3 covers

| Lecture | Topic                | You should be able to                                               |
|---------|----------------------|---------------------------------------------------------------------|
| 18      | Latches              | trace SR, S'R' and D latches; transparency                          |
| 19      | Flip-flops, analysis | characteristic equations; state tables and diagrams; Mealy vs Moore |
| 19      | Timing               | setup, hold, $f_{max}$                                              |
| 20      | Reduction, design    | implication table; D, JK and T design; unused states                |
| 21      | Registers            | parallel load; shift registers; serial adder; ripple counters       |
| 22      | Counters             | synchronous, up-down, BCD, arbitrary, ring, Johnson                 |

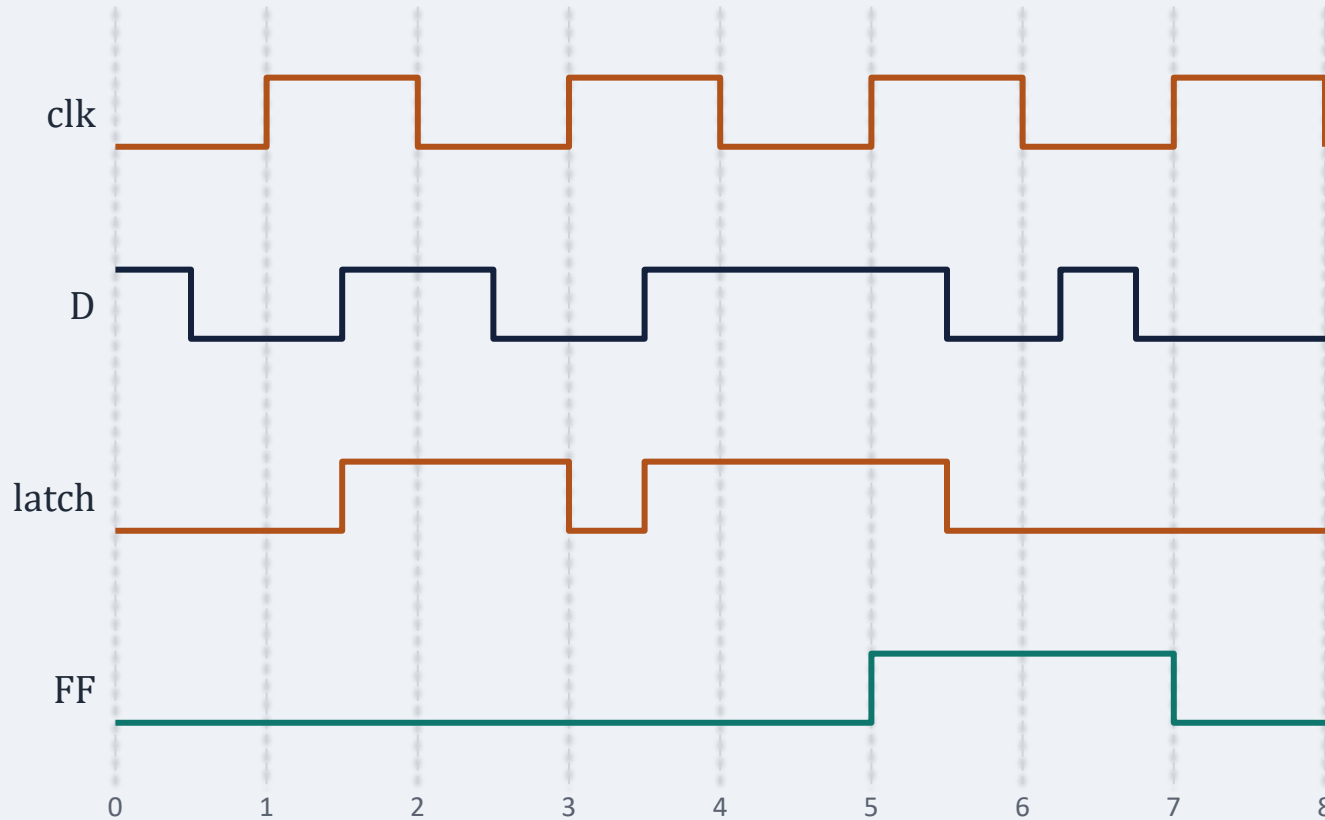
Memorize:  $Q^+ = D$ ,  $Q^+ = JQ' + K'Q$ ,  $Q^+ = T \oplus Q$  and the JK and T excitation tables.

# Review 1 — Latch versus flip-flop

§5.4

## REVIEW

Draw  $Q$  for a D latch (enable = clk) and a positive-edge D flip-flop. Both start at 0.



- 1 FF: samples  $D$  at rising edges 1, 3, 5, 7
- 2 Samples 0, 0, 1, 0  $\rightarrow$  rises at 5, falls at 7
- 3 Latch:  $Q = D$  while  $\text{clk} = 1$ ; holds while  $\text{clk} = 0$
- 4 1–2:  $D$  rises at 1.5  $\rightarrow Q = 1$
- 5 3–4: passes  $D = 0$  at 3, then 1 again at 3.5
- 6 5–6:  $D$  falls at 5.5  $\rightarrow Q = 0$
- 7 The 6.25 pulse comes while  $\text{clk} = 0$ : missed
- 8 7–8:  $D = 0 \rightarrow Q$  stays 0

# Review 2 — JK trace and conversion

§5.4

## REVIEW

A JK flip-flop starts at  $Q = 1$ . Apply  $JK = 01, 11, 11, 00, 10$ . Then make a T flip-flop from a D flip-flop.

| Edge | J K | Action | Q |
|------|-----|--------|---|
| 1    | 0 1 | reset  | 0 |
| 2    | 1 1 | toggle | 1 |
| 3    | 1 1 | toggle | 0 |
| 4    | 0 0 | hold   | 0 |
| 5    | 1 0 | set    | 1 |

- 1 Rules: 00 hold, 01 reset, 10 set, 11 toggle
- 2 Start  $Q = 1$ . Edge 1: 01  $\rightarrow$  reset:  $Q = 0$
- 3 Edge 2: 11  $\rightarrow$  toggle: 1; edge 3: 11  $\rightarrow$  toggle: 0
- 4 Edge 4: 00  $\rightarrow$  hold: 0; edge 5: 10  $\rightarrow$  set: 1
- 5  $Q = 0, 1, 0, 0, 1$
- 6 T from D: we want  $Q^+ = T \oplus Q$ ; a D gives  $Q^+ = D$
- 7 Equate them:  $D = T \oplus Q$ , one XOR fed from its own  $Q$

# Review 3 — Maximum clock frequency

§5.4

## REVIEW

$t_{pd} = 6$ ,  $t_{setup} = 4$ ,  $t_{hold} = 2$  ns. Longest logic path 18 ns, shortest 1 ns. Find  $f_{max}$ ; check hold.

### The two constraints

$$T \geq t_{pd} + t_{comb,max} + t_{setup}$$

$$t_{pd} + t_{comb,min} \geq t_{hold}$$

Setup sets the speed; hold is independent of the clock period.

- 1 Setup:  $T \geq t_{pd} + t_{comb,max} + t_{setup}$
- 2  $= 6 + 18 + 4 = 28$  ns
- 3  $f_{max} = 1/28$  ns = **35.7 MHz**
- 4 Hold: earliest change  $t_{pd} + t_{comb,min} = 6 + 1 = 7$  ns
- 5  $7 \geq 2$  ✓; margin  $7 - 2 = 5$  ns
- 6 Hold fails only if  $t_{pd} + t_{min} < 2$  ns
- 7 A hold violation cannot be fixed by slowing the clock

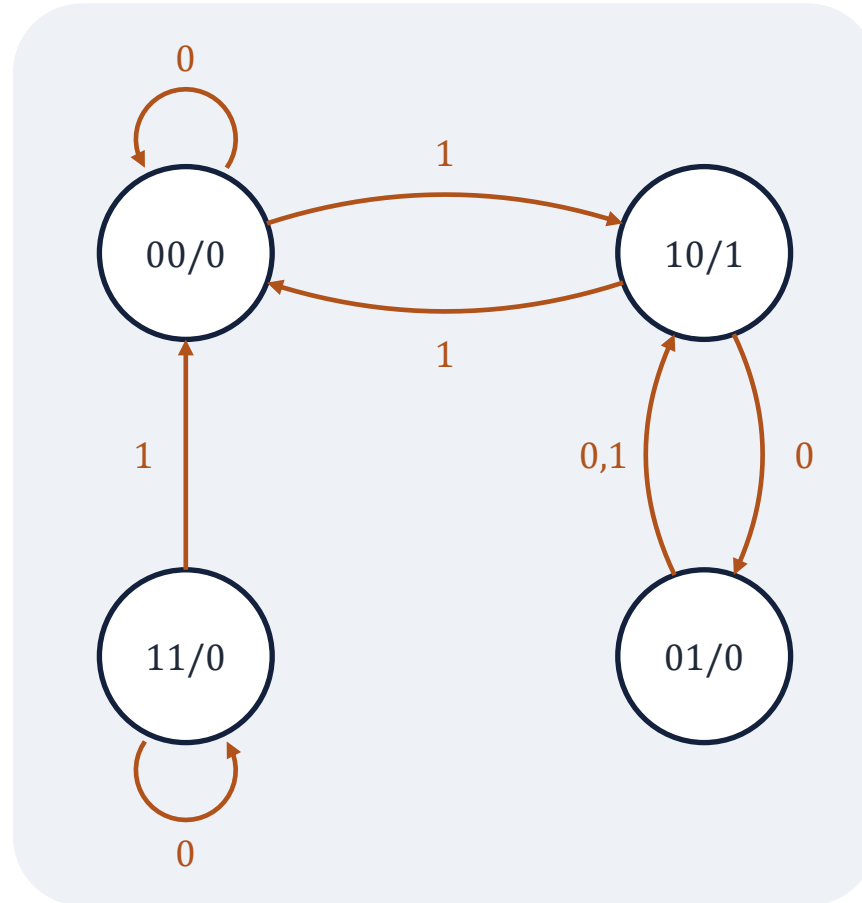
# Review 4 — Analyze a circuit

§5.5

## REVIEW

$D_A = A'x + Bx'$ ,  $D_B = Ax'$ ,  $y = AB'$ . Find the state table and state diagram. Mealy or Moore?

| A B | x | A <sup>+</sup> B <sup>+</sup> | y |
|-----|---|-------------------------------|---|
| 0 0 | 0 | 0 0                           | 0 |
| 0 0 | 1 | 1 0                           | 0 |
| 0 1 | 0 | 1 0                           | 0 |
| 0 1 | 1 | 1 0                           | 0 |
| 1 0 | 0 | 0 1                           | 1 |
| 1 0 | 1 | 0 0                           | 1 |
| 1 1 | 0 | 1 1                           | 0 |
| 1 1 | 1 | 0 0                           | 0 |



- 1 D flip-flops:  $A^+ = D_A$ ,  $B^+ = D_B$
- 2  $A^+ = A'x + Bx'$
- 3  $B^+ = Ax'$
- 4 Each row: put  $A, B, x$  into both
- 5 e.g. 10,  $x = 0$ :  $A^+ = 0$ ,  $B^+ = 1 \rightarrow 01$
- 6  $y = AB'$  uses the state only: **Moore**
- 7  $y = 1$  only in state 10

# Break

10 minutes — we resume at the 0:55 mark

Next: state reduction, design, registers and counters

# Review 5 — Reduce the states

§5.7

## REVIEW

Reduce this state table (next state and output for  $x = 0$ ,  $x = 1$ ).

| State | Next $x=0$ | Next $x=1$ | $y$ $x=0$ | $y$ $x=1$ |
|-------|------------|------------|-----------|-----------|
| a     | b          | c          | 0         | 0         |
| b     | d          | e          | 0         | 0         |
| c     | f          | e          | 0         | 0         |
| d     | a          | e          | 0         | 1         |
| e     | a          | e          | 0         | 1         |
| f     | a          | e          | 0         | 1         |

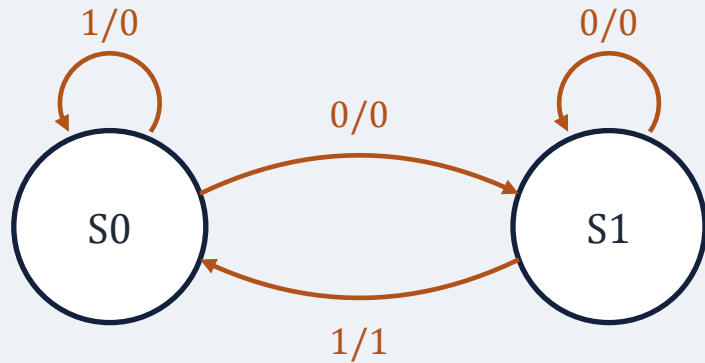
- 1 Group by outputs:  $\{a, b, c\}$  give 0, 0;  $\{d, e, f\}$  give 0, 1
- 2  $d, e, f$  all go to  $(a, e)$ : identical  $\rightarrow d \equiv e \equiv f$
- 3 Replace  $e$  and  $f$  by  $d$  everywhere
- 4 Now  $b \rightarrow (d, d)$  and  $c \rightarrow (d, d)$ : identical  $\rightarrow b \equiv c$
- 5  $a \rightarrow (b, b)$  differs from  $b \rightarrow (d, d)$ :  $a$  stays alone
- 6 Classes  $\{a\}, \{b, c\}, \{d, e, f\}$ : **six states  $\rightarrow$  three**
- 7 Two flip-flops instead of three

# Review 6 — Design a detector

§5.8

## REVIEW

Mealy machine:  $y = 1$  when the input sequence ends in 01. Use one D flip-flop.



$S_1$  = 'the last input was 0'.

| A | x | $A^+$ | y |
|---|---|-------|---|
| 0 | 0 | 1     | 0 |
| 0 | 1 | 0     | 0 |
| 1 | 0 | 1     | 0 |
| 1 | 1 | 0     | 1 |

- 1  $S_1$ : the last input was 0
- 2 Codes:  $S_0 = 0, S_1 = 1$  (one flip-flop A)
- 3 From any state:  $x = 0 \rightarrow S_1; x = 1 \rightarrow S_0$
- 4 So  $A^+ = x'$ :  $D_A = x'$
- 5  $y = 1$  in  $S_1$  with  $x = 1$  (ends in 01)
- 6  $y = Ax$
- 7 Mealy:  $y$  uses  $x$

# Review 7 — Shift register trace

§6.2

## REVIEW

A shift-right register holds 0110. The serial input into the MSB is 1, 1, 0, 1. Trace it.

| Clock | SI | $A_3A_2A_1A_0$ |
|-------|----|----------------|
| 0     | —  | 0110           |
| 1     | 1  | 1011           |
| 2     | 1  | 1101           |
| 3     | 0  | 0110           |
| 4     | 1  | 1011           |

- 1 Every clock: bits move right; SI enters  $A_3$ ;  $A_0$  is lost
- 2 Clock 1, SI = 1: 0110 → 1011
- 3 Clock 2, SI = 1: 1011 → 1101
- 4 Clock 3, SI = 0: 1101 → 0110
- 5 Clock 4, SI = 1: 0110 → **1011**
- 6 After 4 clocks it holds the 4 serial bits; the oldest is in  $A_0$

# Review 8 — Counters

§6.4

## REVIEW

(a) Build a mod-6 counter from a 4-bit counter with synchronous load. (b) Compare a 4-bit ring and Johnson counter.

### (a) Mod 6: 0 ... 5

At 5 = 0101, assert Load with inputs 0000.

$Load = Q_2 Q_0$  is enough: among 0–5 only 5 has both bits.

### (b) Ring vs Johnson

Ring: 4 states, one-hot, no decoder.

Johnson: 8 states, 2-input AND to decode each.

- 1 (a) Mod 6 counts 0–5, then back to 0
- 2 At 5 = 0101, load 0000 on the next edge
- 3 Load must be 1 only at 5: only 5 has  $Q_2 = Q_0 = 1$
- 4 So  $Load = Q_2 Q_0$ , a 2-input AND
- 5 Synchronous load acts at the edge: no glitch
- 6 (b) Ring feeds back  $Q_0$ ; Johnson feeds back  $Q_0'$
- 7 4 bits: ring **4** states, Johnson **8** states

# Practice set — 15 minutes

§5.8

YOUR TURN

≈ 15 min

1.  $t_{pd} = 5$ ,  $t_{comb} = 10$ ,  $t_{setup} = 5$  ns. Maximum clock frequency?
2. A JK flip-flop has  $J = K = 1$  and  $Q = 0$ . What is  $Q$  after one edge? After two?
3. Write the excitation table of a T flip-flop.
4. How many flip-flops does a 12-state machine need (binary)?
5. Which counter has no false counts: ripple or synchronous? Why?

Work alone, as on the exam.

# Practice set — Solutions

§5.8

## SOLUTION

1.  $T \geq t_{pd} + t_{comb} + t_{setup} = 5 + 10 + 5 = 20 \text{ ns}$
2.  $f_{max} = 1/20 \text{ ns} = \mathbf{50 \text{ MHz}}$
3.  $J = K = 1$  toggles:  $Q = 0 \rightarrow \mathbf{1}$ ; next edge  $\rightarrow \mathbf{0}$
4.  $T = Q \oplus Q^+$ :  $0 \rightarrow 0:0$ ;  $0 \rightarrow 1:1$
5.  $1 \rightarrow 0:1$ ;  $1 \rightarrow 1:0$
6.  $n$  flip-flops give  $2^n$  states:  $2^3 = 8 < 12 \leq 16 = 2^4$
7. So **4 flip-flops**
8. 5. **Synchronous**: every bit changes on the same edge
9. Ripple: bits change one after another  $\rightarrow$  false counts

# Key takeaways

D

$$Q^+ = D$$

JK

$$Q^+ = JQ' + K'Q$$

T

$$Q^+ = T \oplus Q$$

Clock

$$T \geq t_{pd} + t_{comb} + t_{su}$$

Design

diagram  $\rightarrow$  table  $\rightarrow$  maps

Counters

check unused states

Sequential design is a procedure: follow it every time.

Homework #11 due at the start of Lecture 28

Next: Exam 3, then random-access memory